

Set Theory

Mathematics for Computing Assignment

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1 Introduction

Set theory underlies much of modern mathematics and computer science, giving a precise vocabulary for talking about collections of objects. A **set** is simply a well-defined collection of distinct objects, called its **elements**. Sets are conventionally named with capital letters, and their elements are listed inside curly braces.

Example. Let $S = \{2, 3, 5, 7, 11\}$. Here S is the set of the first five prime numbers, and each of 2, 3, 5, 7, 11 is an element of S .

2 Types of Sets

2.1 Cardinality of a Set

The **cardinality** of a set is the number of elements it contains, written $|A|$.

Example. If $A = \{a, e, i, o, u\}$ then $|A| = 5$, since there are five vowels listed.

2.2 Finite Set

A set is **finite** if its elements can be completely counted.

Example. $A = \{10, 20, 30, 40\}$ is finite, with $|A| = 4$.

2.3 Infinite Set

A set is **infinite** if it does not terminate — it has no last element.

Example. $\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$, the set of integers, is infinite.

2.4 Empty (Null) Set

The **empty set**, written $\emptyset$ or $\{\}$, has no elements at all.

Example. The set of even prime numbers greater than 2 is $\emptyset$, since 2 is the only even prime.

2.5 Singleton Set

A **singleton set** contains exactly one element.

Example. $A = \{7\}$ is a singleton set, so $|A| = 1$.

2.6 Equal Sets

Two sets are **equal** if they contain precisely the same elements, irrespective of order or repetition.

Example. $A = \{p, q, r\}$ and $B = \{r, q, p\}$ satisfy $A = B$.

2.7 Equivalent Sets

Two sets are **equivalent** if they share the same cardinality, even when their elements are different.

Example. $A = \{1, 2, 3\}$ and $B = \{cat, dog, bird\}$ are equivalent, since $|A| = |B| = 3$.

2.8 Sets Have No Duplicates

By definition, a set never records repeated elements more than once.

Example. $\{4, 4, 5, 5, 5, 6\} = \{4, 5, 6\}$.

2.9 Subset

A is a **subset** of B , written $A \subseteq B$, if every element of A is also an element of B .

Example. $A = \{2, 4\}$, $B = \{1, 2, 3, 4, 5\} \Rightarrow A \subseteq B$.

2.10 Proper Subset

If $A \subseteq B$ and $A \neq B$, then A is a **proper subset** of B , written $A \subset B$.

Example. $A = \{1, 2\}$, $B = \{1, 2, 3\} \Rightarrow A \subset B$.

2.11 Superset

If $A \subseteq B$, then B is called a **superset** of A , written $B \supseteq A$.

Example. $B = \{1, 2, 3, 4\}$, $A = \{3, 4\} \Rightarrow B \supseteq A$.

2.12 Universal Set

The **universal set** U contains every element relevant to a given discussion.

Example. If we are studying digits, $U = \{0, 1, 2, \dots, 9\}$ and any digit set considered is a subset of U .

2.13 Power Set

The **power set** $P(A)$ is the set of all subsets of A , including $\emptyset$ and A itself. If $|A| = n$, then $|P(A)| = 2^n$.

Example. Let $A = \{x, y\}$. Then

$$P(A) = \{\emptyset, \{x\}, \{y\}, \{x, y\}\}, \quad |P(A)| = 2^2 = 4.$$

2.14 Countable Set

A set is **countable** if its elements can be listed in a sequence matching the natural numbers, $1, 2, 3, \dots$

Example. The set of integers $\mathbb{Z}$ is countably infinite, even though it extends in both directions.

2.15 Uncountable Set

A set is **uncountable** if no such listing is possible.

Example. The set of real numbers $\mathbb{R}$, or any interval such as $(0, 1)$, is uncountable.

3 Operations on Sets

Throughout this section, take

$$A = \{2, 3, 4, 5\}, \quad B = \{4, 5, 6, 7\}$$

as the running example.

3.1 Union of Sets

The **union** $A \cup B$ collects every element found in A , in B , or in both.

$$A \cup B = \{x : x \in A \text{ or } x \in B\}$$

For the sets above, $A \cup B = \{2, 3, 4, 5, 6, 7\}$.

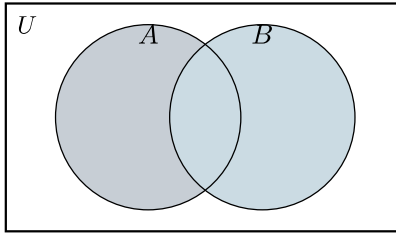


Figure 1: $A \cup B$ (shaded region)

3.2 Intersection of Sets

The **intersection** $A \cap B$ keeps only the elements shared by both sets.

$$A \cap B = \{x : x \in A \text{ and } x \in B\}$$

For the sets above, $A \cap B = \{4, 5\}$.

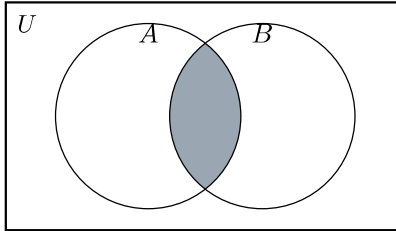


Figure 2: $A \cap B$ (shaded region)

3.3 Difference of Sets

The **difference** $A - B$ consists of elements in A that are not in B .

$$A - B = \{x : x \in A, x \notin B\}$$

For the sets above, $A - B = \{2, 3\}$ and, by contrast, $B - A = \{6, 7\}$ — the operation is not symmetric.

3.4 Symmetric Difference

The **symmetric difference** $A \triangle B$ contains elements in exactly one of the two sets.

$$A \triangle B = (A - B) \cup (B - A)$$

For the sets above, $A \triangle B = \{2, 3, 6, 7\}$.

3.5 Complement of a Set

Relative to a universal set U , the **complement** A^c consists of every element of U not in A .

$$A^c = U - A$$

Example. Let $U = \{1, 2, \dots, 9\}$ and $A = \{2, 3, 4, 5\}$. Then $A^c = \{1, 6, 7, 8, 9\}$.

3.6 Cartesian Product

The **Cartesian product** $A \times B$ is the set of all ordered pairs formed by taking one element from A and one from B .

$$A \times B = \{(a, b) : a \in A, b \in B\}, \quad |A \times B| = |A| \cdot |B|$$

Example. Let $A = \{p, q\}$, $B = \{1, 2\}$. Then

$$A \times B = \{(p, 1), (p, 2), (q, 1), (q, 2)\}, \quad |A \times B| = 2 \times 2 = 4.$$

4 Principle of Inclusion and Exclusion

This principle gives a systematic way to count the elements in a union of sets without over-counting elements that lie in more than one of them.

4.1 Two Sets

$$|A \cup B| = |A| + |B| - |A \cap B|$$

The intersection is subtracted once because simply adding $|A|$ and $|B|$ counts common elements twice.

4.2 Three Sets

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|$$

4.3 Worked Example

Suppose in a class of students, $|A| = 18$ study Python, $|B| = 12$ study Java, and $|A \cap B| = 6$ study both. The number who study at least one of the two languages is

$$|A \cup B| = 18 + 12 - 6 = 24.$$

5 Partially Ordered Sets (Posets)

A **partially ordered set**, or **poset**, is a set P together with a relation $\preceq$ that is:

1. **Reflexive** — $a \preceq a$ for every $a \in P$;
2. **Antisymmetric** — if $a \preceq b$ and $b \preceq a$, then $a = b$;
3. **Transitive** — if $a \preceq b$ and $b \preceq c$, then $a \preceq c$.

A poset is written $(P, \preceq)$.

5.1 Example

Consider the power set of $\{p, q\}$,

$$P(\{p, q\}) = \{\emptyset, \{p\}, \{q\}, \{p, q\}\},$$

ordered by the subset relation $\subseteq$. This forms a poset $(P(\{p, q\}), \subseteq)$.

5.2 Hasse Diagram

A Hasse diagram represents a poset by drawing an edge upward from a to b whenever $a \preceq b$ and there is no element strictly between them.

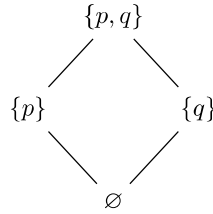


Figure 3: Hasse diagram of $P(\{p, q\})$

Here $\emptyset$ is the smallest element and $\{p, q\}$ is the largest, with $\{p\}$ and $\{q\}$ incomparable to each other.

6 Conclusion

Set theory supplies the basic language for reasoning about collections of objects, and its operations — union, intersection, difference, symmetric difference, complement, and Cartesian product — recur throughout discrete mathematics and computer science. The Principle of Inclusion and Exclusion extends this toolkit to precise counting problems, while partially ordered sets introduce the idea of structured comparison between elements, a concept that reappears in areas such as lattice theory, database design, and algorithm analysis.

END OF ASSIGNMENT